

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

WEIGHTED ENSEMBLE STEGANALYSIS

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled "Weighted Ensemble Steganalysis: Integrating CNN, XGBoost, and LightGBM for Robust Detection of Hidden Information in Digital Images" in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Isha Kansal. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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Abstract

Digital image steganography is a covert communication technique that enables the embedding of hidden information within ordinary-looking digital images. While this capability serves legitimate purposes such as copyright protection and secure data transmission, it simultaneously poses serious challenges in cybersecurity and digital forensics owing to its potential for misuse. The detection of such concealed data, commonly referred to as steganalysis, is an inherently complex undertaking because modern steganographic embedding techniques introduce only minimal, often imperceptible modifications to image structures.

Traditional steganalysis approaches depend on handcrafted statistical features derived from spatial or transform domains. Although such methods are interpretable and moderately effective, they demonstrate limited adaptability in detecting sophisticated and evolving steganographic techniques. In contrast, deep learning approaches, particularly Convolutional Neural Networks (CNNs), have demonstrated a strong ability to learn hierarchical and discriminative features directly from raw image data, offering superior performance in identifying subtle pixel-level anomalies. However, these models are often constrained by overfitting, high computational demands, and heavy reliance on large, labelled datasets.

To overcome these competing limitations, this project implements a Weighted Heterogeneous Ensemble Model (WHEM) that integrates the complementary strengths of both traditional and deep learning paradigms. The proposed model combines three distinct components: a specialized steganalysis CNN inspired by GAN discriminators for residual noise learning, alongside two gradient boosting models, XGBoost and LightGBM, for statistical feature analysis. A central contribution of this work is the implementation of an optimized weight-search mechanism that identifies the most effective combination of model outputs by maximizing the F1-score on a validation set.

Experimental results on a simulated and balanced dataset of 5,000 images demonstrate that the ensemble model achieves superior classification performance compared to all individual models, attaining an overall accuracy of approximately 74.50%. The findings affirm that CNNs effectively capture noise-based patterns, while gradient boosting models detect statistical irregularities, and their weighted integration leads to enhanced detection reliability and robustness.

Keywords: *Steganalysis, Digital Steganography, Convolutional Neural Networks, XGBoost, LightGBM, Ensemble Learning, Heterogeneous Ensemble, Weighted Averaging, Digital Forensics, Cybersecurity*

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1. Introduction

1.1 Background and Motivation

In the digital era, the exponential growth of internet-based communication has made the secure and private transmission of information more critical than ever before. Among the diverse techniques employed to ensure data confidentiality, digital image steganography occupies a unique and significant position. Unlike encryption, which transforms the content of a message into an unreadable format, steganography conceals the very existence of the message by embedding it within a carrier medium, most commonly a digital image. This fundamental distinction renders steganography particularly powerful as a covert communication tool and, at the same time, an instrument of serious concern for cybersecurity professionals.

The practical applications of steganography span a wide range of legitimate uses, including digital watermarking, intellectual property protection, and covert data transmission in high-risk environments. However, the same characteristics that make it useful for legitimate purposes also make it an attractive tool for malicious actors engaged in cybercrime, industrial espionage, and unauthorized data exfiltration. Detecting such hidden data, a discipline known as steganalysis, is therefore a task of paramount importance in the domains of digital forensics and information security.

1.2 The Steganalysis Problem

Steganalysis is the science of detecting hidden information within digital media. The task is inherently complex because modern steganographic techniques are specifically designed to minimize perceptible distortion and replicate the statistical properties of natural images. As a consequence, detecting the presence of embedded data becomes a binary classification challenge: distinguishing between cover images, which contain no hidden data, and stego images, which have had data embedded within them.

Historically, steganalysis relied upon handcrafted statistical features extracted from image data. Techniques such as co-occurrence matrices, higher-order pixel statistics, and the widely adopted Spatial Rich Model (SRM) proved effective in capturing the subtle noise patterns and statistical anomalies introduced by the embedding process. While these methods offered interpretability and provided valuable forensic insights, they were fundamentally constrained by their dependence on predefined feature sets and their inability to generalize to the increasingly adaptive and sophisticated steganographic methods that emerged over time.

The advent of deep learning revolutionized the field. Convolutional Neural Networks demonstrated the ability to learn discriminative features end-to-end, directly from raw pixel data, eliminating the need for manual feature engineering. Architectures such as XuNet and YedidiaNet introduced fixed high-pass filtering in their initial layers to suppress image content and amplify steganographic artifacts, achieving state-of-the-art detection performance. Nevertheless, CNN-based approaches are not without their own limitations: susceptibility to overfitting on small

datasets, high computational resource requirements, and a tendency to inadvertently learn image content rather than focusing exclusively on steganographic residuals.

1.3 Scope and Objectives

Against this backdrop, this project undertakes the design, implementation, and evaluation of a Weighted Heterogeneous Ensemble Model (WHEM) for digital image steganalysis. The project brings together three fundamentally different modelling approaches, a specialized steganalysis CNN, XGBoost, and LightGBM, into a unified classification framework governed by an optimized weight-search mechanism. The primary objectives of this project are as follows:

- To develop a balanced dataset of cover and stego images using multiple embedding techniques, reflecting real-world steganographic diversity.
- To design and train a CNN architecture capable of learning residual noise representations from high-pass filtered images.
- To extract and leverage statistical features from images for training gradient boosting classifiers.
- To implement a weighted ensemble fusion mechanism with systematic weight optimization based on F1-score maximization.
- To evaluate the ensemble model against individual baselines using accuracy, precision, recall, F1-score, and ROC-AUC metrics.

1.4 Relevance of the Problem

The significance of robust steganalysis extends well beyond academic interest. As digital communication networks grow in scale and sophistication, the potential for covert channels to be used in organized cybercrime, terrorism financing, and intellectual property theft demands that detection systems remain both adaptive and reliable. Ensemble learning approaches, particularly heterogeneous ensembles that integrate diverse feature representations, represent a promising frontier for addressing these real-world demands.

This project addresses several critical gaps identified in the existing literature. Few studies have effectively integrated fundamentally different model types, such as deep learning networks and statistical gradient boosting classifiers, into a unified framework for steganalysis. Furthermore, most existing ensemble methods rely on fixed or heuristic weight assignments rather than systematic optimization, limiting the potential performance gains from model combination. The proposed WHEM addresses these shortcomings directly.

2. Methodology

2.1 Overview of the WHEM Pipeline

The proposed methodology introduces a Weighted Heterogeneous Ensemble Model (WHEM) designed to improve steganalysis performance by integrating multiple complementary learning approaches within a structured pipeline. The overall pipeline encompasses five major stages: dataset construction and preprocessing, statistical feature extraction, individual model training, ensemble fusion with weight optimization, and final evaluation.

This multi-stage design ensures that each component of the model is trained on the most appropriate representation of the image data. Deep learning components operate on residual noise maps derived from high-pass filtered images, while machine learning components operate on handcrafted statistical feature vectors. The integration of these fundamentally different representations within a unified probabilistic framework constitutes the defining innovation of the proposed approach.

2.2 Dataset Construction

The first stage of the methodology involves the construction of a balanced and representative dataset. A dataset comprising 5,000 images, divided equally between 2,500 cover images and 2,500 stego images, was assembled to ensure unbiased training and evaluation. All images were standardized to a uniform resolution of 256 x 256 pixels and converted to grayscale to facilitate consistent processing across models.

Stego images were generated using multiple embedding strategies, including Least Significant Bit (LSB) substitution and adaptive embedding methods. This deliberate variety in embedding technique ensures that the dataset reflects the diversity of real-world steganographic scenarios, where adversaries may employ strategies of varying complexity and payload capacity. The dataset was subsequently partitioned into training, validation, and test subsets following a 70-15-15 split, ensuring sufficient data for robust model training and unbiased evaluation.

2.3 Preprocessing

The preprocessing stage was designed to enhance the visibility of steganographic artifacts while suppressing irrelevant image content that might confound classification. For CNN-based processing, high-pass filters inspired by the Spatial Rich Model (SRM) were applied to each image to generate residual noise maps. These residual maps highlight the subtle pixel-level modifications introduced during the data embedding process, making them ideally suited as inputs to the CNN, whose architecture is specifically designed to focus on noise residuals rather than semantic image content.

For machine learning model processing, a separate preprocessing pipeline was implemented. Statistical features were extracted from the original and residual images, including co-occurrence matrices, entropy measures, pixel difference histograms, and higher-order statistical descriptors. Feature scaling procedures, including normalization and standardization, were applied to all

extracted features to ensure numerical stability and consistent convergence during gradient boosting model training.

2.4 Model Architecture

The WHEM integrates three independent predictive components, each bringing a distinct form of representational capacity to the ensemble.

The CNN component was specifically designed for steganalysis. Inspired by GAN discriminator architectures, the network begins with a fixed high-pass filtering layer that suppresses image content while amplifying residual noise. Subsequent convolutional layers, combined with batch normalization and activation functions, learn increasingly abstract representations of steganographic artifacts from these residuals.

The XGBoost component was trained on the extracted statistical feature vectors. XGBoost employs a gradient-boosted ensemble of decision trees with strong regularization capabilities, effectively learning complex nonlinear relationships in high-dimensional statistical feature spaces while controlling overfitting.

The LightGBM component was also trained on the statistical feature vectors. LightGBM employs histogram-based decision tree learning and parallel processing, offering faster training and competitive predictive performance. Its leaf-wise tree growth strategy often produces more accurate splits than the level-wise approach used by XGBoost, introducing valuable diversity into the ensemble.

2.5 Ensemble Fusion and Weight Optimization

The central and most innovative component of the methodology is the ensemble fusion mechanism. Each of the three trained models produces a probability score for each input image, indicating the likelihood of it being a stego image. These individual probability outputs are combined through a weighted averaging scheme expressed as follows:

$$P_{\text{final}} = w_1 \times P_{\text{CNN}} + w_2 \times P_{\text{XGB}} + w_3 \times P_{\text{LightGBM}}$$

where the weights w_1 , w_2 , and w_3 are non-negative real numbers constrained to sum to one. The proposed approach employs a systematic grid-search-based optimization procedure to identify the optimal weight combination by maximizing the ensemble F1-score on the validation set. This ensures that each model contributes to the final prediction in proportion to its relative predictive strength.

2.6 Evaluation Methodology

The evaluation methodology was designed to provide a comprehensive and multi-faceted assessment of model performance. The primary metrics employed were accuracy, precision, recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC). These metrics collectively assess not only the overall classification accuracy but also the balance between false positive and false negative rates, which is of particular importance in forensic applications where both types of error carry significant consequences.

3. Tools and Technologies

3.1 Development Environment

The project was developed entirely within a Python-based software environment, leveraging a suite of widely adopted open-source libraries for deep learning, machine learning, data processing, and evaluation. The choice of tools was governed by their established reliability, community support, and suitability for the tasks at hand.

3.2 Deep Learning Framework: TensorFlow / Keras

The CNN component of the WHEM was implemented using TensorFlow in conjunction with the Keras high-level API. TensorFlow provides the computational backend for defining, compiling, and training neural network architectures, with support for GPU-accelerated computation. Keras simplified the construction of the CNN architecture, including the definition of the fixed high-pass filtering layer, convolutional blocks, batch normalization layers, and the final classification head.

3.3 Gradient Boosting Libraries: XGBoost and LightGBM

XGBoost was used for training the first gradient boosting component. This library implements an optimized and scalable version of gradient-boosted decision trees, incorporating both L1 and L2 regularization to mitigate overfitting. Its support for parallel processing and efficient memory usage makes it well-suited to working with high-dimensional statistical feature vectors.

LightGBM was employed for the second gradient boosting component. The library implements a histogram-based learning approach that reduces memory footprint and accelerates training, making it particularly efficient for large feature sets.

3.4 Machine Learning Utilities: Scikit-learn

The Scikit-learn library provided a unified interface for preprocessing, feature scaling, model evaluation, and grid search operations. Specifically, it was used for implementing the normalization and standardization of statistical feature vectors, computing performance metrics, generating confusion matrices, and conducting the hyperparameter tuning grid search for the ensemble weight optimization procedure.

3.5 Programming Language and Environment

Python served as the primary programming language for all components of the project. Its extensive ecosystem of scientific computing libraries, combined with its clean and readable syntax, made it the most appropriate choice for a project integrating components from deep learning, classical machine learning, and statistical analysis.

4. Implementation

4.1 Experimental Configuration

The experimental setup was designed to evaluate the effectiveness of the proposed WHEM under controlled and reproducible conditions. A simulated dataset of 5,000 images was used, equally balanced between cover and stego classes to prevent class imbalance from biasing the model. All images were resized to a uniform resolution of 256 x 256 pixels and processed in grayscale. The dataset was partitioned into training, validation, and test sets following a 70-15-15 ratio. The configuration details of all major components are summarized in Table 2 below.

Component	Configuration
Dataset Size	5,000 images (2,500 cover, 2,500 stego)
Image Resolution	256 x 256 pixels (grayscale)
Train/Validation/Test Split	70 / 15 / 15
CNN Training Epochs	40 to 50
CNN Batch Size	32
CNN Optimizer	Adam (adaptive learning rate)
XGBoost Estimators	300
LightGBM Estimators	400
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, AUC

Table 1: Experimental Setup and Hyperparameter Configuration

4.2 Preprocessing Implementation

All input images were loaded and converted to grayscale before being subjected to high-pass filtering. The high-pass filtering step was implemented as a fixed, non-trainable layer within the CNN, using filter kernels derived from the Spatial Rich Model. This approach ensured that the CNN operated exclusively on residual noise images, suppressing scene content while preserving the subtle embedding artifacts that constitute the steganalysis signal.

For the gradient boosting models, a separate feature extraction pipeline was executed. Co-occurrence matrices, entropy-based descriptors, pixel difference histograms, and higher-order statistical moments were computed for each image. The resulting feature vectors were then subject to min-max normalization and z-score standardization to ensure that all features contributed on a comparable scale during model training.

4.3 Model Training

The CNN was trained over 40 to 50 epochs using the Adam optimizer with a batch size of 32. Early stopping was employed to prevent overfitting, monitoring validation loss during training. The architecture included convolutional layers with increasing filter depths, interspersed with batch normalization and non-linear activation functions, concluding with fully connected layers and a sigmoid output unit for binary classification.

XGBoost was trained on the extracted statistical feature matrix with 300 estimators. Hyperparameters including maximum tree depth, learning rate, and subsample ratio were tuned through grid search on the validation set. LightGBM was similarly trained using 400 estimators with histogram-based optimization. Both models were cross-validated to assess generalization and guard against overfitting.

4.4 Ensemble Weight Optimization

Following individual model training, the ensemble fusion layer was implemented. Predicted probability outputs from the CNN, XGBoost, and LightGBM on the validation set were collected. A grid search over the weight space was performed, systematically varying the contribution of each model's probability output across all feasible combinations. For each weight triplet, the ensemble F1-score on the validation set was computed. The weight combination yielding the highest F1-score was selected as the final ensemble configuration and subsequently applied to the held-out test set for unbiased performance evaluation.

5. Major Findings / Results

5.1 Individual Model Performance

The performance of each individual model component was assessed independently on the held-out test set prior to ensemble construction. Among the three models, XGBoost achieved the highest standalone accuracy at approximately 72.50%, reflecting the effectiveness of statistical feature-based learning for capturing the global distributional changes introduced by steganographic embedding. LightGBM followed closely with an accuracy of approximately 72.25%, demonstrating comparable discriminative capability.

The CNN model achieved a notably lower standalone accuracy of approximately 55.75%. This outcome is consistent with the well-documented finding that CNN-based models require large volumes of training data to reach their full discriminative potential. Importantly, the CNN contributes features that are fundamentally different in nature from those captured by the gradient boosting models, making it a valuable and complementary participant in the ensemble.

Model	Accuracy
XGBoost	0.7250
LightGBM	0.7225
Steganalysis CNN	0.5575

Table 2: Individual Model Performance on Test Set

5.2 Ensemble Model Performance

The WHEM ensemble significantly outperformed all three individual models, achieving a test set accuracy of approximately 74.50%. This improvement of approximately two percentage points over the best individual model (XGBoost at 72.50%) underscores the tangible value of combining complementary feature representations within a weighted ensemble framework.

Class	Precision	Recall	F1-Score	Support
Cover (0)	0.7285	0.7931	0.7594	203
Stego (1)	0.7654	0.6954	0.7287	197
Accuracy	-	-	0.7450	400
Macro Avg	0.7469	0.7443	0.7441	400
Weighted Avg	0.7467	0.7450	0.7443	400

Table 3: Classification Report of the WHEM Model

5.3 Confusion Matrix Analysis

The confusion matrix for the ensemble model on the test set revealed the following classification outcomes:

	Predicted Cover (0)	Predicted Stego (1)
Actual Cover (0)	161 (True Negative)	42 (False Positive)
Actual Stego (1)	60 (False Negative)	137 (True Positive)

Table 4: Confusion Matrix of WHEM Model (Test Set, n=400)

The confusion matrix reveals a balanced distribution of correct and incorrect classifications across both classes, with no pronounced bias toward either cover or stego predictions. The ensemble model successfully reduced both false positive and false negative rates compared to individual models, enhancing its reliability for real-world forensic deployment.

5.4 ROC-AUC Analysis

The ROC curve analysis further confirmed the superior discriminative capability of the WHEM ensemble. The ensemble achieved a higher AUC score compared to all individual models, reflecting its ability to maintain strong true positive rates across a wide range of classification thresholds. This characteristic is particularly significant in forensic applications, where practitioners may wish to adjust the decision threshold based on the relative operational costs of false positives and false negatives.

5.5 Key Observations and Error Analysis

Several important observations emerged from the results. The ensemble effectively mitigated the specific weaknesses of each individual component: the CNN contributed noise-based local patterns that the gradient boosting models were unable to capture through statistical features alone, while XGBoost and LightGBM contributed global statistical anomaly detection that compensated for the CNN's limited standalone accuracy on the small dataset.

Error analysis revealed that the WHEM model encountered the greatest difficulty with images featuring extremely low embedding payloads and highly textured image content. Despite these inherent challenges, the ensemble substantially reduced misclassification rates relative to individual models, validating the robustness of the heterogeneous integration approach.

6. Conclusion and Future Scope

6.1 Conclusion

This project has presented a novel and practically motivated approach to digital image steganalysis through the design, implementation, and evaluation of a Weighted Heterogeneous Ensemble Model (WHEM). By integrating a specialized steganalysis CNN with two gradient boosting classifiers, XGBoost and LightGBM, within a weight-optimized ensemble framework, the proposed model effectively overcomes the principal limitations of both traditional feature-based methods and standalone deep learning approaches.

The study has demonstrated that combining deep learning's capacity for learning pixel-level noise representations with gradient boosting's effectiveness in capturing global statistical irregularities yields classification performance that surpasses what any single model can achieve in isolation. Experimental validation on a balanced dataset of 5,000 images confirmed that the WHEM achieves an accuracy of approximately 74.50%, outperforming the best individual model by approximately two percentage points.

Beyond performance metrics, this project makes a conceptual contribution by demonstrating that the integration of fundamentally different machine learning paradigms within a unified and optimized framework constitutes a highly effective strategy for complex detection tasks characterized by subtle and diverse signals.

6.2 Future Scope

While the proposed model achieves promising results, several directions for further improvement and expansion have been identified. The use of larger and more diverse datasets, including real-world steganographic samples generated with a broader range of contemporary embedding algorithms, would significantly strengthen the model's generalization capability.

The integration of GAN-based steganographic images into the training dataset would further enhance the model's robustness against the most advanced and adaptive embedding techniques currently employed. The incorporation of transformer-based vision architectures represents another promising direction for combining with CNNs and gradient boosting models within an expanded heterogeneous ensemble.

Optimization of the model for real-time steganalysis deployment is also an important area for future work. Techniques such as model pruning, quantization, and knowledge distillation can substantially reduce inference latency while preserving detection accuracy. Enhancing the interpretability of the model through Grad-CAM and SHAP visualizations would make model predictions more transparent and accountable in legal and forensic contexts.

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Appendices

Appendix A: Glossary of Key Terms

Term	Definition
Steganography	The practice of concealing information within a carrier medium (e.g., a digital image) such that the existence of the hidden data is not apparent to observers.
Steganalysis	The discipline concerned with detecting the presence of hidden data embedded within digital media through steganographic techniques.
Cover Image	An unmodified digital image that contains no hidden embedded information.
Stego Image	A digital image that has been modified to contain hidden embedded information through a steganographic process.
CNN	A class of deep learning model particularly well-suited to processing structured grid data such as images, capable of learning hierarchical feature representations end-to-end.
XGBoost	An optimized, scalable gradient-boosted decision tree ensemble algorithm incorporating regularization techniques to prevent overfitting.
LightGBM	A gradient boosting framework that employs histogram-based learning and leaf-wise tree growth for efficient and accurate classification.
WHEM	Weighted Heterogeneous Ensemble Model; the proposed model integrating CNN, XGBoost, and LightGBM with optimized weight fusion.

SRM	Spatial Rich Model; a high-dimensional feature extraction framework for steganalysis employing linear and nonlinear filters to capture statistical dependencies.
ROC-AUC	Receiver Operating Characteristic - Area Under the Curve; a metric assessing classifier discrimination performance across all classification thresholds.
LSB	Least Significant Bit; a basic steganographic embedding technique replacing the least significant bits of pixel values with secret data bits.

Appendix B: Notation Summary

Symbol	Description
P_CNN	Probability score output by the CNN model for a given image
P_XGB	Probability score output by the XGBoost model for a given image
P_LightGBM	Probability score output by the LightGBM model for a given image
P_final	Final ensemble probability score (weighted average of individual model outputs)
w1, w2, w3	Ensemble weights assigned to CNN, XGBoost, and LightGBM respectively (sum to 1)
F1-Score	Harmonic mean of precision and recall; primary metric for weight optimization
AUC	Area Under the ROC Curve; measure of overall discriminative ability